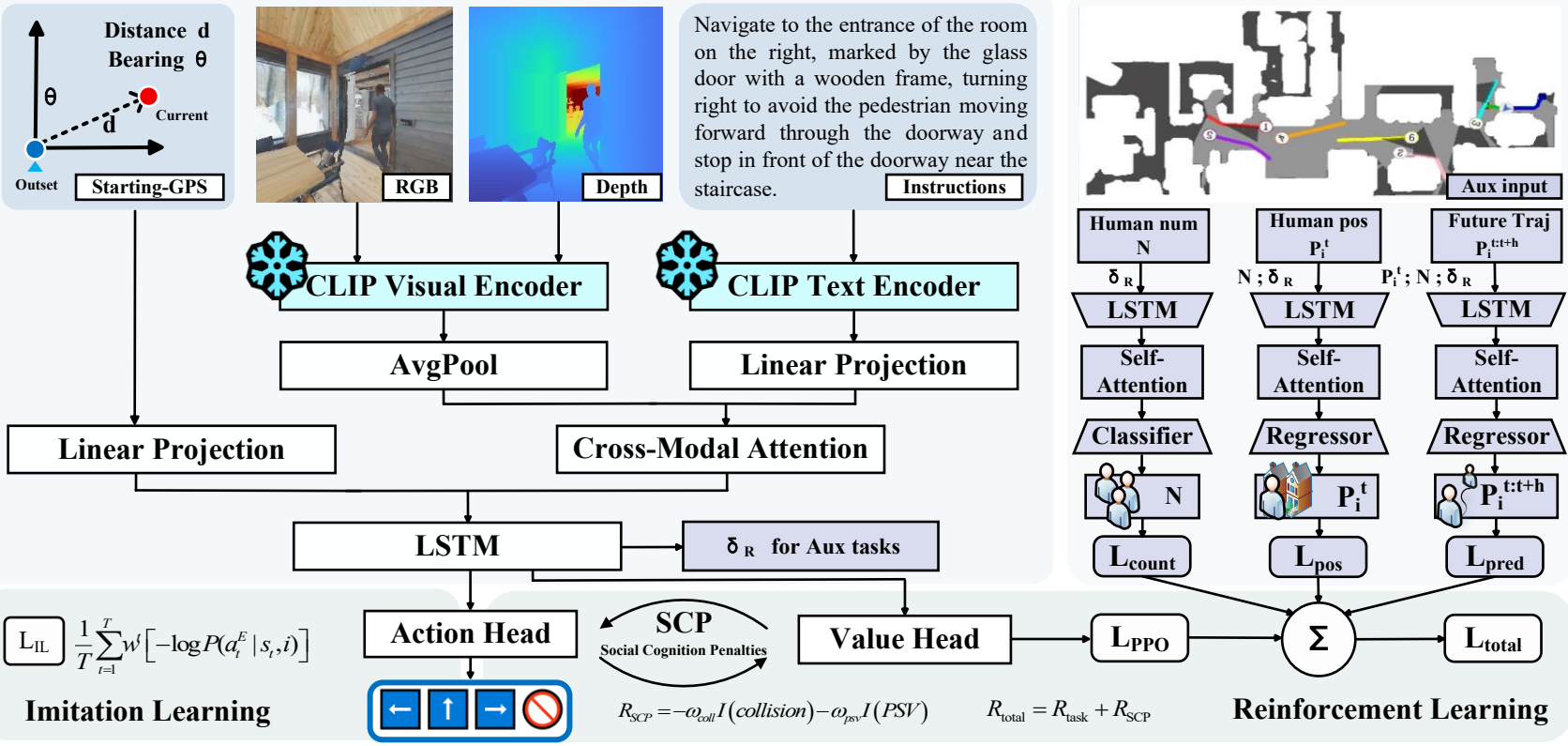

Distance d
Bearing θ
θ
d
Current
Outset
Starting-GPS
RGB
Depth
Navigate to the entrance of the room on the right, marked by the glass door with a wooden frame, turning right to avoid the pedestrian moving forward through the doorway and stop in front of the doorway near the staircase.
Instructions
Aux input
CLIP Visual Encoder
CLIP Text Encoder
AvgPool
Linear Projection
Linear Projection
Cross-Modal Attention
LSTM
δ R for Aux tasks
Human num N
Human pos P_i^t
Future Traj P_i^{t:t+h}
δ R
N ; δ R
P_i^t ; N ; δ R
LSTM
LSTM
LSTM
Self-Attention
Self-Attention
Self-Attention
Classifier
Regressor
Regressor
N
P_i^t
P_i^{t:t+h}
L_count
L_pos
L_pred
L_IL
$\frac{1}{T}\sum_{t=1}^{T} w^t\left[-\log P(a_t^E \mid s_t, i)\right]$
Action Head
SCP
Social Cognition Penalties
Value Head
L_PPO
Σ
L_total
Imitation Learning
$R_{SCP} = -\omega_{coll} I(collision) - \omega_{psv} I(PSV)$
$R_{total} = R_{task} + R_{SCP}$
Reinforcement Learning